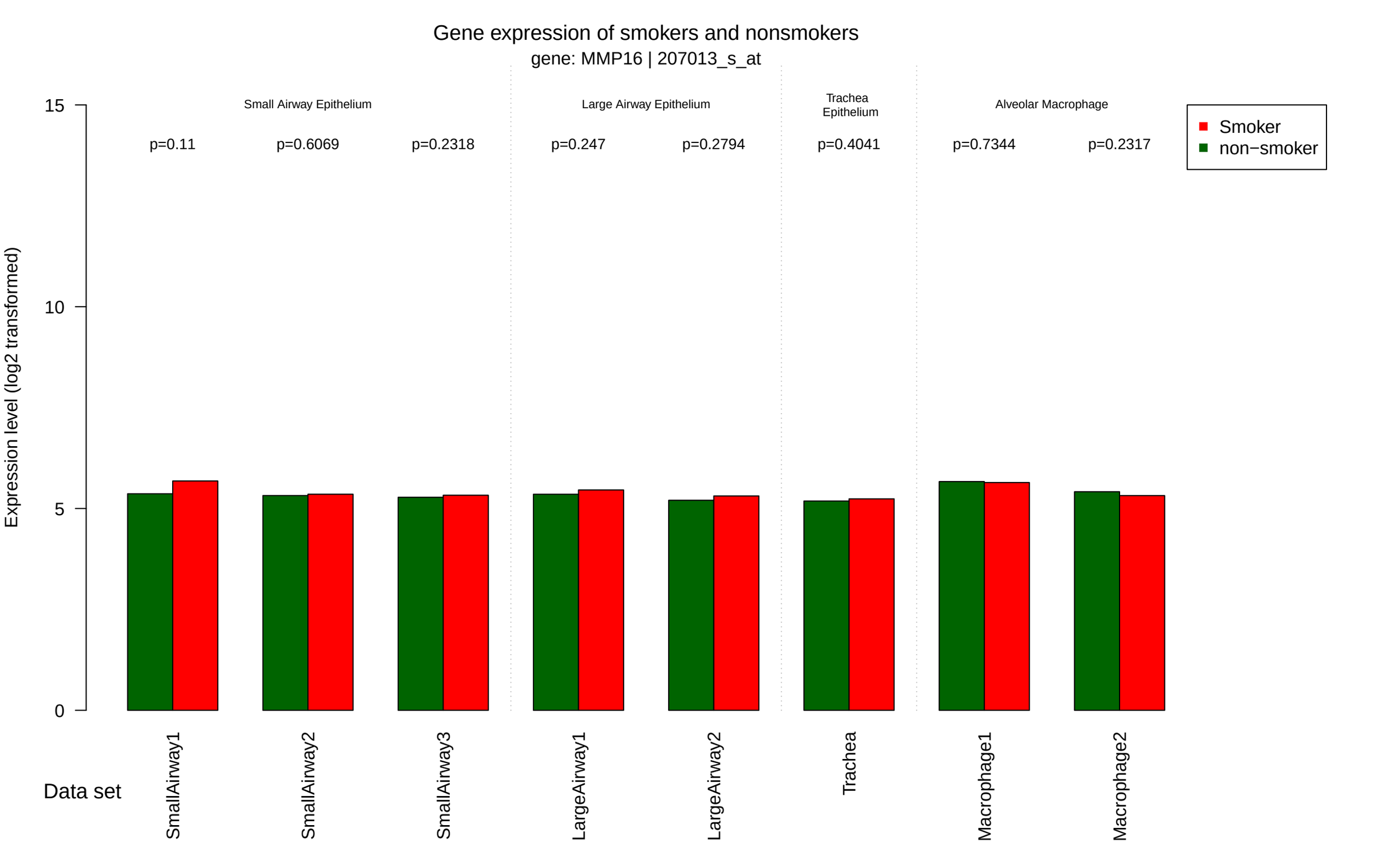
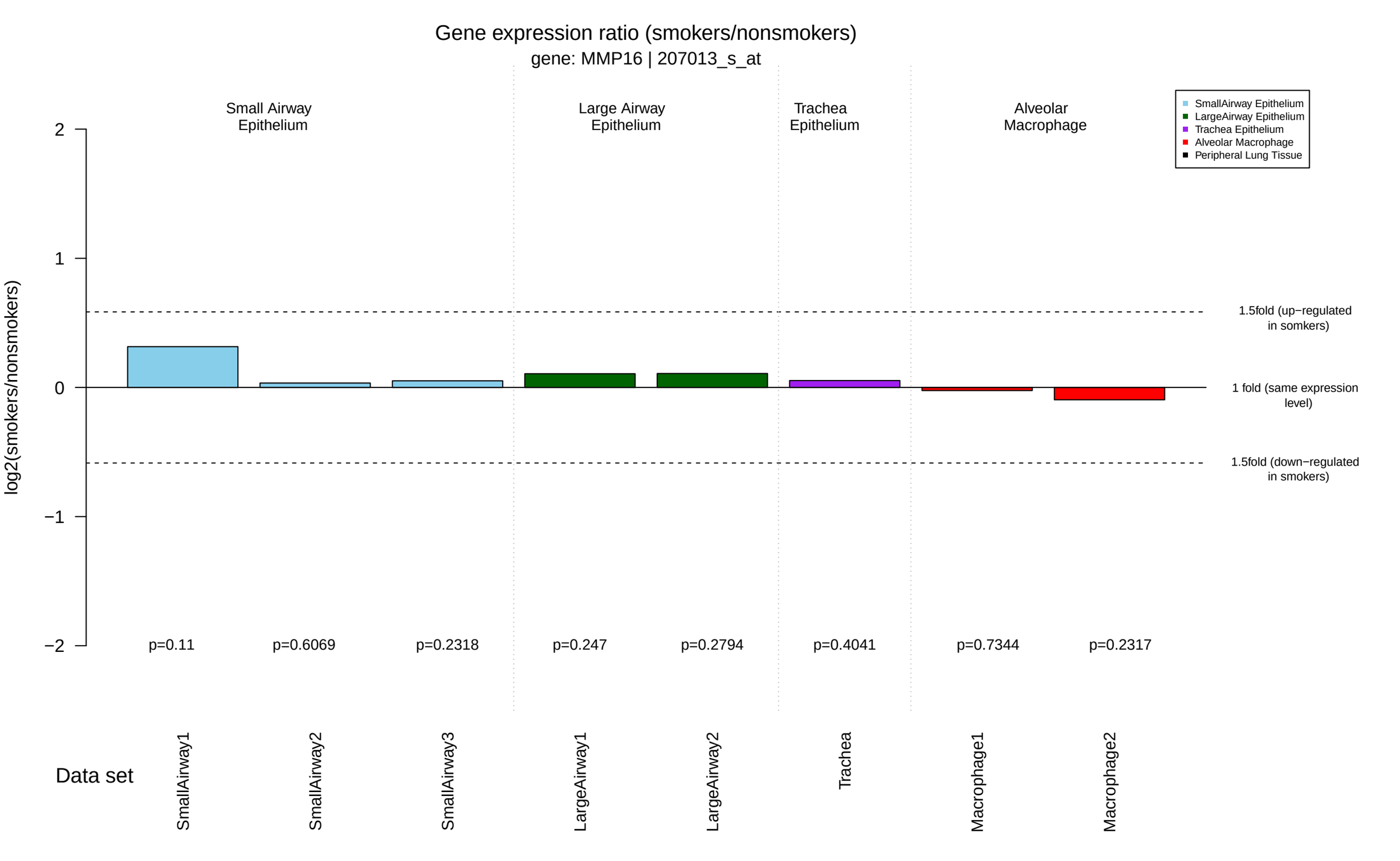
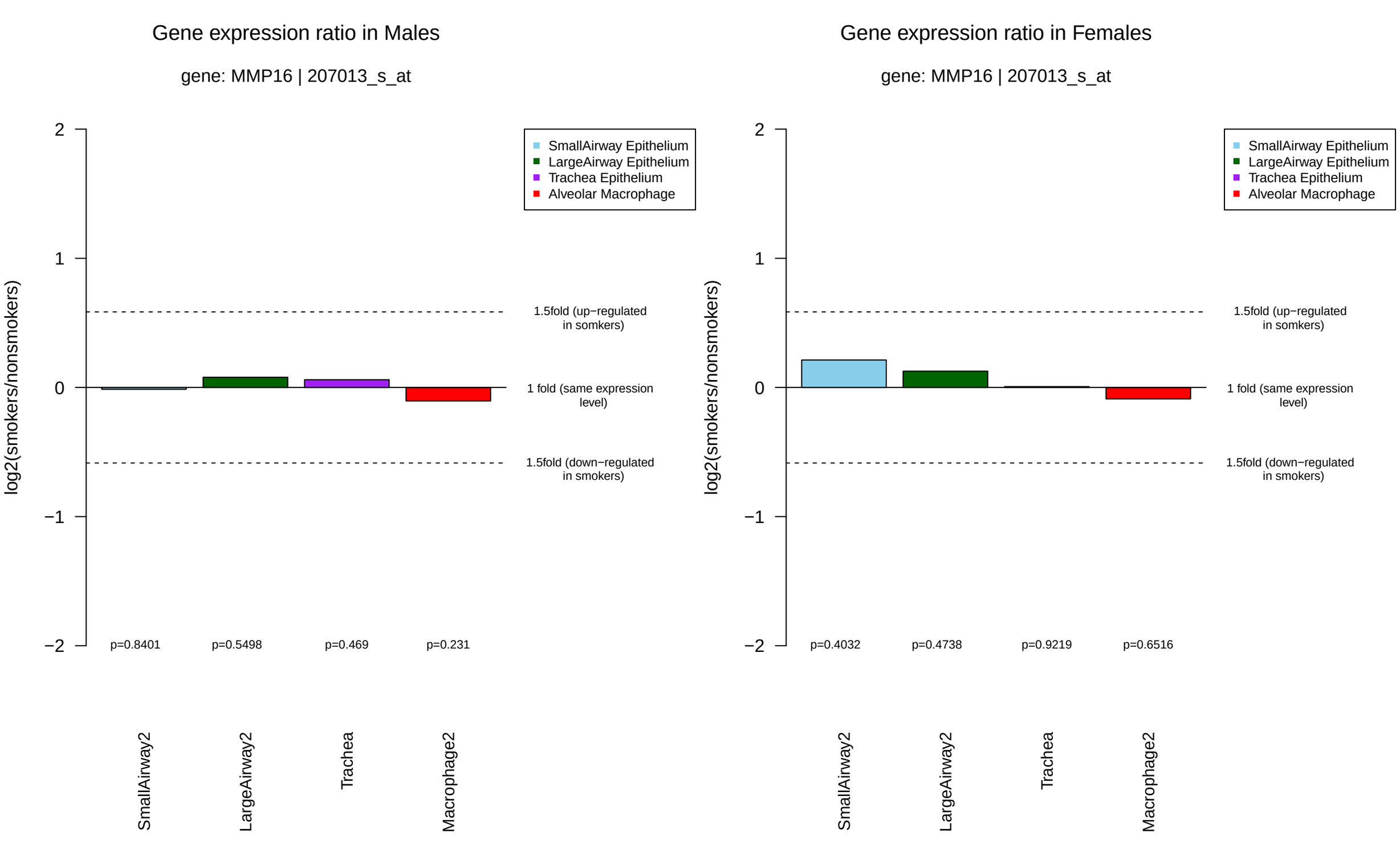
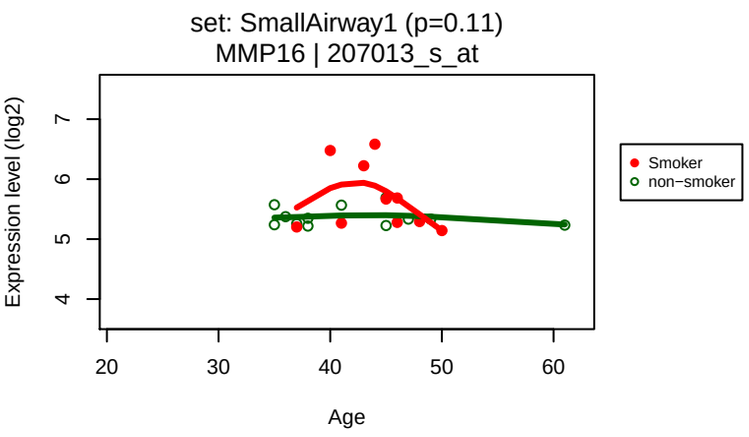
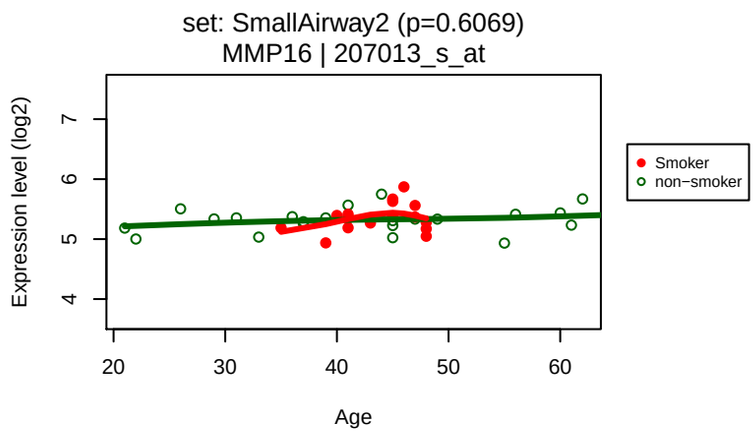
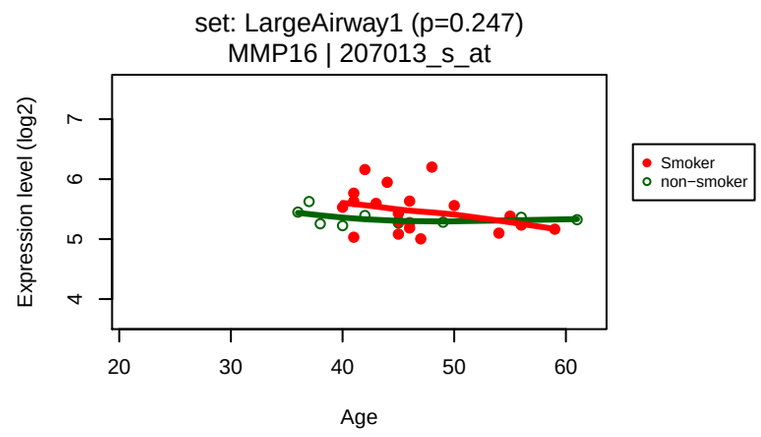

Gene expression of smokers and nonsmokers
gene: MMP16 | 207013_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.11
p=0.6069
p=0.2318
p=0.247
p=0.2794
p=0.4041
p=0.7344
p=0.2317
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

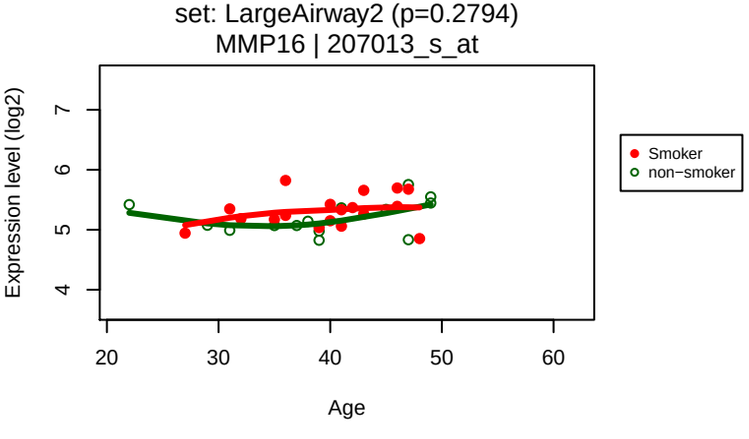

Gene expression ratio (smokers/nonsmokers)
gene: MMP16 | 207013_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.11
p=0.6069
p=0.2318
p=0.247
p=0.2794
p=0.4041
p=0.7344
p=0.2317
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

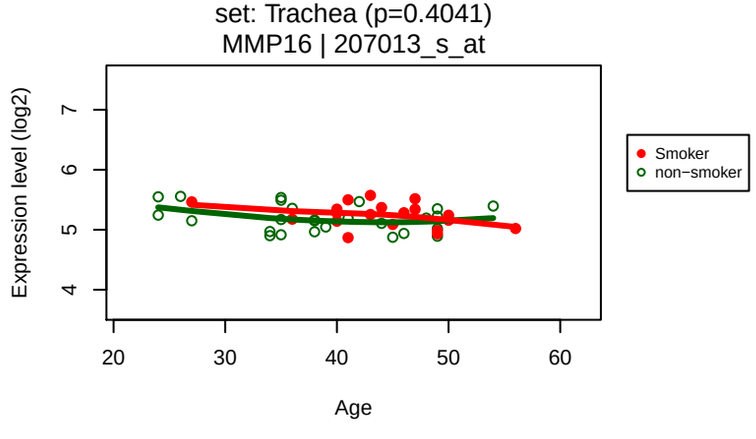

Gene expression ratio in Males
gene: MMP16 | 207013_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8401
p=0.5498
p=0.469
p=0.231
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP16 | 207013_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4032
p=0.4738
p=0.9219
p=0.6516
SmallAirway2
LargeAirway2
Trachea
Macrophage2

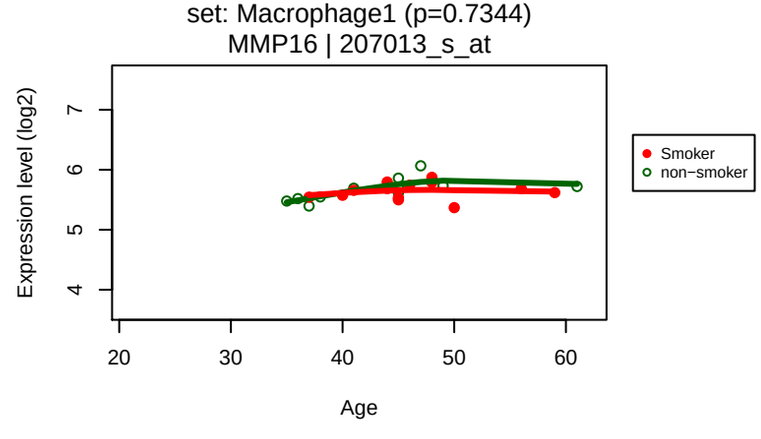

set: SmallAirway1 (p=0.11)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker

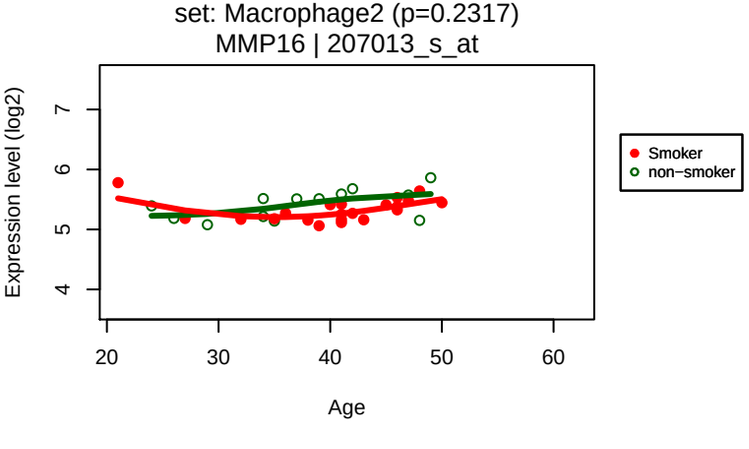

set: SmallAirway2 (p=0.6069)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker

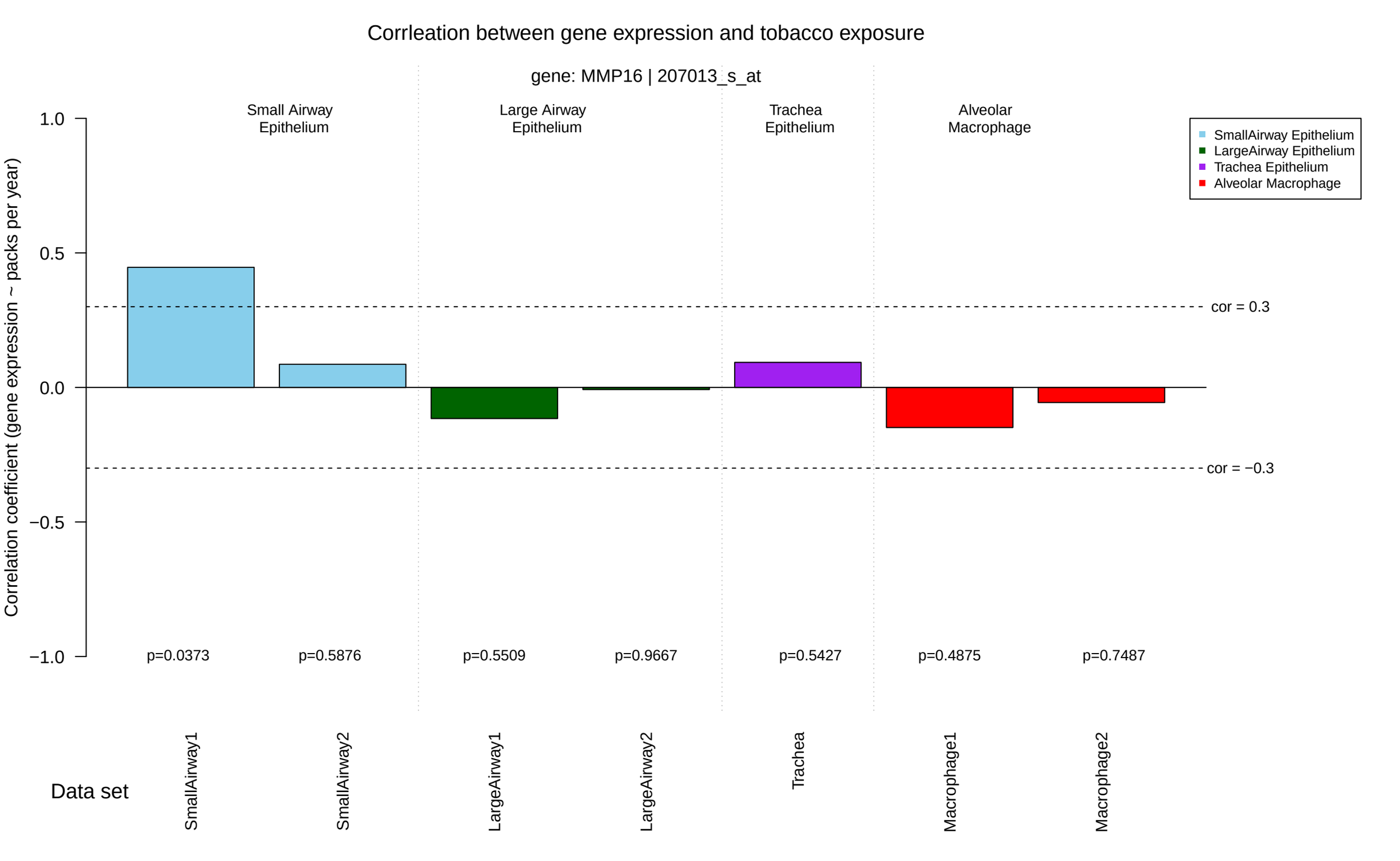

set: LargeAirway1 (p=0.247)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2794)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4041)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7344)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2317)
MMP16 | 207013_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP16 | 207013_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0373
p=0.5876
p=0.5509
p=0.9667
p=0.5427
p=0.4875
p=0.7487
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2